

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A SENARIO BASED QUESTIONS

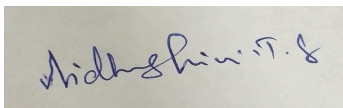
Criteria	Understanding of Scenario (2)	Identification of Key Issues (2)	Application of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I Vidhushini.T.S (192511061) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics (L3, L4)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

- Identify missing values in **Age**, **Monthly_Spend**, and **Purchase_Frequency**
- Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
- Compute the imputed values and justify the selected method

B. Outlier Detection and Treatment

- Detect outliers in **Age** and **Monthly_Spend** using the **Z-score method**
- Identify records containing extreme values (e.g., Age = 105, Spend = 500)
- Apply suitable treatment methods (**capping** / **transformation** / **removal**) and justify

C. Categorical Data Standardization

- Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
- Convert all entries into a standardized format (**Male** / **Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

- i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
- ii. Interpret whether the relationship is **positive or negative**
- iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection

- i. Identify redundant attributes using **correlation or domain knowledge**
- ii. Select an optimal subset of features for modeling
- iii. Justify your selection

C. Data Transformation

- i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
- ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics (L3, L4, L5)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

- Apply **3 bins** for Study_Hours and Test_Score
 - Calculate bin width and define intervals
-

B. Equal-Frequency Binning

- Divide data into **3 bins with equal records**
 - Assign values accordingly
-

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis (*L3, L4, L5*)

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.



Sample Dataset: Visit Duration

Customer_ID Visit_Duration (seconds)

C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600



Tasks

A. Equal-Frequency Binning

- Divide the dataset into **3 bins** with approximately equal number of records
 - Assign each value to its respective bin
 - Analyze how effectively this method groups short and long visit durations
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B. Z-Score Normalization

- Compute the **mean and standard deviation** of visit duration
- Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

- Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**
-

C. Min-Max Normalization

- Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- Transform all values accordingly
- Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Question 1: Data Preprocessing in Retail Analytics.

The retail company wants to prepare customer purchase data for a recommendation system. The dataset contains missing values, inconsistent gender formats and extreme values.

Therefore data preprocessing is required before applying the recommendation model.

A Handling Missing Values:

i) Identification of Missing Values:

Customer	Age	Monthly Spend	Purchase Frequency
C1	23	20	5
C2	Missing	35	8
C3	31	Missing	10
C4	105	500	50
C5	27	25	Missing
C6	40	45	12
C7	36	38	9
C8	42	300	15

Age $\rightarrow$ C2 is missing Monthly Spend $\rightarrow$ C3 is missing

Purchase Frequency $\rightarrow$ C5 is missing

ii) Median Imputation

Age: 23, 27, 31, 36, 40, 42, 105

Median = ~~23~~ $7+1/2 = 4 \Rightarrow 36$ C2 Age = 36

Monthly Spend: 20, 25, 35, 38, 45, 300, 500

Median = $7+1/2 \Rightarrow 4 \Rightarrow 38$

C3 - Monthly - Spend = ~~35~~ 38 K

Purchase Frequency: 5, 8, 9, 10, 12, 45, 50

$$\text{Median} = 7 + 1/2 = 4 \Rightarrow 10$$

C5 Purchase-Frequency = 10

iii) Mean Imputation

$$\text{Age, Mean} = \frac{23 + 31 + 105 + 27 + 40 + 36 + 42}{7} = 43.43$$

$$\text{C2 Age} = 43.43$$

$$\text{Monthly spend, Mean} = \frac{20 + 35 + 500 + 25 + 45 + 38 + 300}{7} = 137.57$$

$$\text{C3 Monthly Spend} = 137.57$$

$$\text{Purchase Frequency, Mean} = \frac{5 + 8 + 10 + 50 + 45 + 12 + 9}{7} = 19.86$$

$$\text{C5 Purchase Frequency} = 19.86$$

B) Outlier in Detection and Treatment

$$\text{Z-Score, } Z = \frac{X - \mu}{\sigma}$$

Normally, a value with an absolute Z-score greater than approximately 3 is considered a strong statistical outlier.

However, because this is a small dataset, extreme values should also be checked using domain knowledge.

Age: The value 105 is clearly unrealistic for ^{normal} small retail customer dataset.

C4 → Age = 105 → Extreme value

Monthly Spend: C4 → ₹500k C8 → ₹300k are much higher than most other customers.

Suitable Treatment: I would use capping (winorization) instead of immediately deleting the records.

C. Categorical Data Standardization:

The Gender column contains:

M, Male, male, F, Female, FEMALE.

Standardized Values:

Original	Standardized
M	Male
Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

Therefore, the final categories should contain only:
Male / Female.

Question : 2 - Covariance & Data Reduction in E-commerce

Analysis.

The e-commerce company has several financial and behavioral attributes. Some of these attributes are correlated, causing redundancy and increasing computational ^{-ional} cost. The objective is therefore to identify relationship between attributes and select useful features.

A. Covariance Analysis:

Calculate the covariance between Income and Purchase - Value.

Customer	Income	Purchase Value
U1	40	100
U2	60	150
U3	85	210
U4	120	320
U5	45	110
U6	75	180
U7	150	360
U8	70	160

Calculate Means:

$$\text{Mean (Income)} = \frac{40 + 60 + 85 + 120 + 45 + 75 + 150 + 70}{8} = 80.625$$

$$\text{Mean (Purchase)} = \frac{100 + 150 + 210 + 320 + 110 + 180 + 360 + 160}{8} = 198.75$$

Covariance Formula:

$$\text{Cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

$$\text{Cov (Income, Purchase Value)} \approx 3529.46$$

The positive covariance indicates a positive relationship between the two attributes.

B. Feature Selection:

The dataset contains: Age, ^{Income}~~contains~~, Purchase-Value, Credit-Score, Orders, Avg-Balance, EMI

Some features may contain overlapping information.

Selected Features:

Income, Purchase-value, Credit-Score, Orders and Avg-Balance.

Age can be removed if it does not contribute significantly to customer classification.

Highly correlated features should also be examined using correlated analysis before final selection.

Justification:

1. Reduces data dimensionally.
2. Removes unnecessary information
3. Reduces computational cost
4. Reduces redundancy
5. Can improve model performance
6. Makes the predictive model easier to interpret.

C. Data Transformation:

The numerical attributes should be standardized

The standardization formula is:

$$Z = \frac{X - \mu}{\sigma}$$

For example, income values ranging from 40 to 150, while EMI has values ranging from 5000 to 18000.

Attributes to standardize:

Age, Income, Purchase-Value, Credit-Score, Orders, Avg-Balance, EMI.

Cust-ID and Customer-class should not be standardize because they are an identifier and categorical class respectively.

Question 2: Data Transformation & Discretization in Education Analytics.

The education analytics team wants to convert continuous values such as study hours and test scores into ^{-us} meaningful groups. This improves interpretation and decision-making.

A. Equal-Width Binning

It divides the entire range into intervals having equal width.

1. Study Hours:

Minimum = 2 Maximum = 11 No. of bins = 3

$$\text{Bin Width} = \frac{11 - 2}{3} = 3.$$

Study Hours	Bin
2, 3, 4	Bin 1
5, 6, 7	Bin 2
8, 9, 10, 11	Bin 3

2. Test Score:

Minimum = 40 Maximum = 100

$$\text{Bin Width} = \frac{100 - 40}{3} = 20.$$

Test Score	Bin
40, 50, 55	Bin 1
65, 75	Bin 2
85, 90, 95, 98, 100	Bin 3

B. Equal - Frequency Binning:

It divides dataset so that each bin contains approx. same number of records.

3+3+4 records.

Study Hours

Bin	Values
Bin 1	2, 3, 4
Bin 2	5, 6, 7
Bin 3	8, 9, 10, 11

Test Scores

Bin	Values
Bin 1	40, 50, 55
Bin 2	65, 75, 85
Bin 3	90, 95, 98, 100

C. Concept Hierarchy Construction:

The continuous variables can be converted into meaningful categories.

Study Hours:

Low: 2-4 hrs

Medium: 5-7 hrs

High: 8-11 hrs

Test Score:

Poor: 40-55

Average: 65-85

Excellent: 90-100.

Question: 4 - Normalization of Data:

Given Data: 200, 300, 400, 600, 1000

(a) Min-Max Normalization.

$$\text{Formula: } X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

$$X_{\min} = 200$$

$$X_{\max} = 1000$$

$$X' = \frac{X - 200}{800}$$

$$\text{For } 200 = \frac{200 - 200}{800} = 0$$

$$\text{For } 300 = \frac{300 - 200}{800} = 0.125$$

$$\text{For } 400 = \frac{400 - 200}{800} = 0.25$$

$$\text{For } 600 = \frac{600 - 200}{800} = 0.50$$

$$\text{For } 1000 = \frac{1000 - 200}{800} = 1$$

Final Answer:

0, 0.125, 0.25, 0.50, 1

(b) Z-Score Normalization:

Formula: $Z = \frac{x - \mu}{\sigma}$

$$\text{Mean} = \mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = 500.$$

$$\sigma \approx 282.84$$

x	Z-Score
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

Question-5: Data Transformation in Online Shopping Behavior Analysis.

The e-commerce company ^{mea}~~ens~~ures customer visit ~~of~~ duration. The values range from 15 sec to 3600 sec, so the data is highly skewed. Transformation and normalization are therefore required before clustering.

Given: 15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600.

A. Equal-Frequency Binning:

There are 10 records and 3 bins.

3+3+4 records.

Bin	Visit Duration	Category
Bin1	15, 30, 45	Short
Bin2	60, 120, 300	Medium
Bin3	600, 900, 1800, 3600	Long

Equal-frequency binning is useful because it distributes customers into groups with approximately equal numbers of records.

B. Z-Score Normalization:

$$Z = \frac{x - \mu}{\sigma}$$

$$\text{Mean: } \mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10}$$

$$\mu = 747$$

$$\text{Standard Deviation: } \sigma \approx 1090.92$$

Z-scores:

Visit Duration	Z-score
15	-0.67
30	-0.66
45	-0.64
60	-0.63
120	-0.57
300	-0.41
600	-0.13
900	-0.14
1800	0.97
3600	2.62

Effect of Extreme Values:

The values 1800 and 3600 sec are ^much larger than the other values.

Z-Score normalization does not remove these extreme values. eg. $Z_{3600} \approx 2.62$.

c. Min - Max Normalization

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

$$X_{\min} = 15$$

$$X_{\max} = 3600$$

$$X' = \frac{X - 15}{3585}$$

Visit Duration	Min-Max Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.079
600	0.163
900	0.251
1800	0.498
3600	1.000

For this dataset, Z-score normalization is useful for understanding how far each visit duration is from the average, while Min-Max normalization is useful when clustering requires all values to lie within a fixed range such as 0-1.